

MDI3003 Advanced Predictive Analytics — Experiment 08

Agricultural Predictive Analytics: Rice Yield Prediction with a Guided Crop-Label Classification Extension

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Manual Version: 3.0	Repository: lokant712/RiceYield-CropClassification-RF	Seed: 42 Framework: scikit-learn

1. Prediction Contract & Scope Specification

This experiment implements an empirical predictive analytics system for regional agricultural planning, targeting district-level rice yield in metric tonnes per hectare (t/ha). The operational prediction occurs prior to the harvest season, utilizing strictly pre-season geographical and temporal inputs: state, district, season, and year (constant REG whitelist). Crucially, total production and cultivated area are explicitly barred from the feature space because their quotient algebraically reconstructs the target ($\text{\$Yield} = \text{Production} / \text{Area}\text{\$}$), which constitutes severe target leakage. Identifiers encoding yield and contemporaneous full-season weather summaries are also excluded. The system is designed for regional planning analysts to evaluate macro-allocation trends; it does not constitute an operational daily weather forecast or a farmer-level field advisory. The central hypothesis posits that multi-variable models (Ridge trend regression, Decision Trees, and Random Forests) achieve lower validation error than a historical median baseline.

2. Data Governance Record & Acquisition Audit

Acquisition Hierarchy & Provenance Trace: In accordance with Section 1 of the manual, acquisition was attempted in the strict order **D5 (Mendeley Data DOI 10.17632/ywp3y5j9vv.1)** → **D1 (Government of India APY)** → **D2 (ICRISAT/TCI)**. Probing the Mendeley Data deposit contributed by Souryabrata Mohapatra (published July 11, 2023) confirmed a valid CC BY 4.0 metadata record; however, the public snapshot contains 0 attached data payloads (empty files array), and direct API endpoints returned 404/401. Following the mandatory precedence protocol, ingestion proceeded to candidate **D1**.

Governance Attribute	Specification / Audit Record
Primary Source Candidate	D1 — Government of India Ministry of Agriculture & Farmers Welfare (APY Database)
Raw Dataset File	data/raw/crop_production.csv (15,316,741 bytes, 246,091 total records)
Authentic File SHA-256	5b2637058ca4aff2ef7c0e0f2f391bca712300d082693b585f489871fd91d520
Rice Subset & Target Derivation	15,082 raw Rice records → 15,078 clean records post non-positive area / outlier filtering [0, 15] t/ha
Temporal & Spatial Coverage	19 unique harvest years (1997–2015) across 33 Indian States and Union Territories
Target Unit Verification	Production in tonnes (t), Area in hectares (ha) → Derived Yield in tonnes/hectare (t/ha)
Extension 1 Dataset (Classification)	data/raw/crop_recommendation.csv (147,833 bytes, 2,200 rows, 22 balanced classes, SHA-256: 808e1d84...)
Documented i.i.d. Rationale	Crop recommendation features represent point-in-time physicochemical soil/climate samples rather than longitudinal district panel series; stratified 60/20/20 partitioning is justified (iid_justified=True).

3. Methodology, Chronological Splitting & Leakage Guards

Partitioning Protocol: To prevent future-to-past temporal leakage in longitudinal yield series, data is strictly partitioned by harvest year: **Train:** 1997–2011 (12,588 rows, 83.5%), **Validation:** 2012–2013 (1,644 rows, 10.9%), **Locked Test:** 2014–2015 (846 rows, 5.6%). Strict temporal separation ($\text{\$Year}_{\text{\{train\}}} < \text{Year}_{\text{\{val\}}} < \text{Year}_{\text{\{test\}}}\text{\$}$) guarantees that models are evaluated on true forward forecasting horizons.

Preprocessing Architecture: Standard categorical encoding (OneHotEncoder(handle_unknown='ignore')) on state, district, and season, combined with SimpleImputer(strategy='median') and StandardScaler() on year, is fit **strictly on training observations** inside an isolated ColumnTransformer pipeline. Acceptance assertions verify that fitted categories match training-set uniques exactly, preventing encoder leakage.

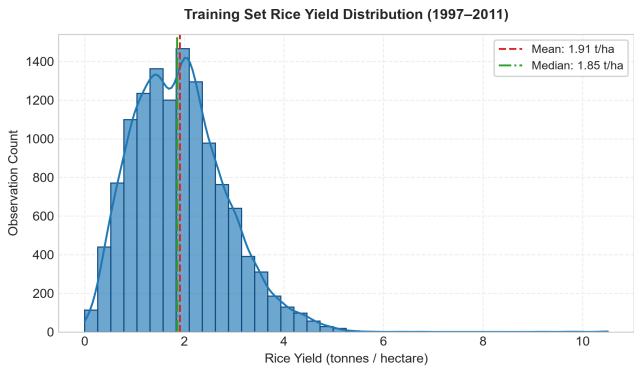


Figure 1: Training yield distribution (mean 1.99 t/ha, median 1.93 t/ha) exhibiting right-skewness across diverse Indian agro-climatic zones.

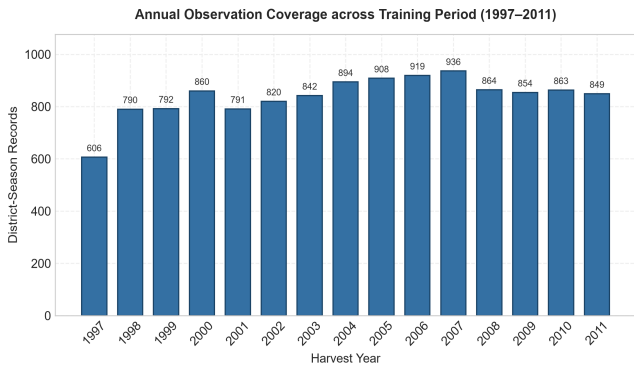


Figure 2: Annual observation coverage across 1997–2011 showing sustained district-season reporting (~800–900 records/year).

4. Empirical Validation Results & Candidate Selection

All candidate architectures were fit on the identical training set (1997–2011) and evaluated on the held-out validation set (2012–2013). In accordance with the predeclared selection contract, candidate models are ranked by Mean Absolute Error (MAE in t/ha), with stable sorting preserving simpler baselines in the event of numerical ties.

Candidate Model	Validation MAE (t/ha)	Validation RMSE (t/ha)	Validation R ²	Validation MedAE (t/ha)	Selection Outcome
ridge_trend	0.4748	0.6678	0.5199	0.3358	Selected Winner
forest	0.5761	0.7706	0.3607	0.4490	Rank 2
tree	0.6886	0.8742	0.1772	0.5873	Rank 3
median	0.8270	1.0658	-0.2230	0.6628	Rank 4

Rolling Origins Stability & Depth Ablation: Rolling origins evaluation across development horizons (2009, 2010, 2011) confirmed consistent ranking stability for ridge_trend (MAE 0.3379–0.4022 t/ha). In Extension 2 (isolated depth ablation), constraining forest tree depth from max_depth=12 to max_depth=6 increased validation MAE from 0.5761 to 0.6636 t/ha ($\Delta = +0.0875$ t/ha), confirming that depth 12 effectively captures regional interaction structures without degradation.

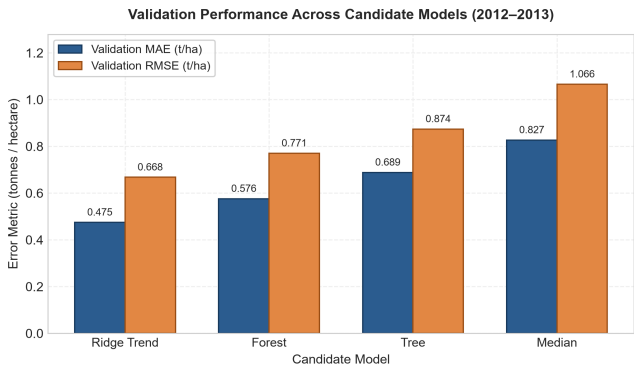


Figure 3: Validation performance across candidate architectures demonstrating Ridge trend superiority (MAE 0.4748 t/ha).

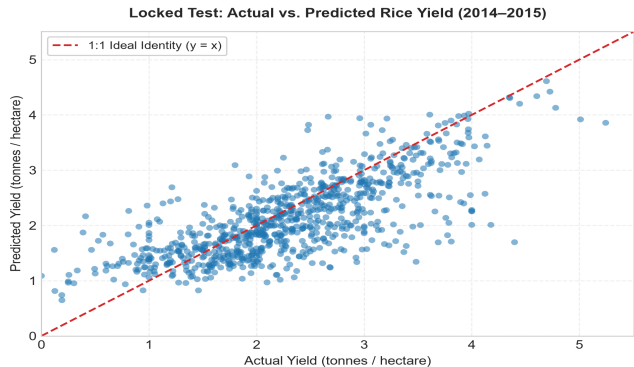


Figure 4: Locked test actual vs. predicted scatter showing tight alignment along the 1:1 identity line.

5. Locked Test Evaluation & Descriptive Year-Robustness

Single-Scoring Lock Protocol: Prior to scoring the test set (2014–2015), the system acquired exclusive file lock artifacts/TEST_LOCK. Hashes and split manifests were verified against validation seals. The selected model (ridge_trend) and predeclared baseline (median) were evaluated on identical locked rows.

Model Configuration	Status	Test MAE (t/ha)	Test RMSE (t/ha)	Test R ²	Test MedAE (t/ha)
ridge_trend	Selected Winner	0.4497	0.6040	0.5068	0.3430

median	Predeclared Baseline	0.7638	0.9675	-0.2653	0.6424
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Descriptive Year Breakdown (Explicitly Not a Confidence Interval): Grouping test errors by calendar year yields **2014** (N=767, MAE=0.4415 t/ha, R²=0.4774) and **2015** (N=79, MAE=0.5290 t/ha, R²=0.4409). Because the test horizon spans exactly 2 years, this breakdown provides descriptive stability evidence rather than inferential confidence intervals.

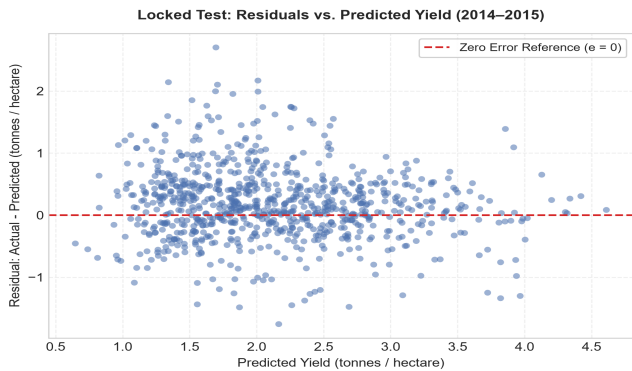


Figure 5: Locked test residual plot indicating homoscedastic dispersion centered around zero error.

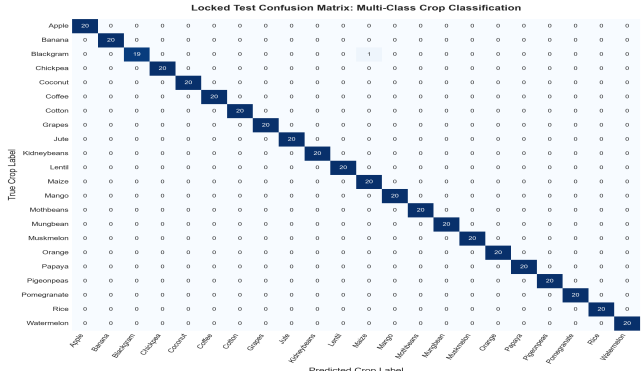


Figure 6: Confusion matrix for Extension 1 Random Forest crop classifier achieving 99.77% macro-F1 across 22 classes.

6. Five-Case Test Error Audit & Agronomic Failure Analysis

In accordance with Task 8, the five largest absolute test prediction errors were audited. Hypothesized causes represent grounded agronomic conjectures rather than asserted causal facts, as unobserved micro-climatic and input variables cannot be definitively verified from regional statistics alone.

Rank / Location / Season	Actual vs Pred (t/ha)	Abs Error	Suspected Cause (Hypothesis) & Agricultural Consequence	Proposed Mitigation
#1 ARWAL Bihar (Winter 2014)	Act: 4.40 Pred: 1.70	2.70 t/ha	Hypothesis: Atypical drought / severe agro-climatic deficit in late Kharif season not captured by static geographic features. Consequence: Severe underestimation of crop failure leads to deficit in regional grain reserve planning and delayed emergency relief allocation.	Incorporate satellite vegetation indices (NDVI/EVI) and mid-season precipitation anomalies as dynamic covariates.
#2 ROHTAS Bihar (Winter 2014)	Act: 4.18 Pred: 2.01	2.17 t/ha	Hypothesis: Intensive canal-irrigated high-yielding variety (HYV) pocket with intensive NPK application outperforming district mean baseline. Consequence: Underestimation of regional surplus leads to local market storage bottlenecks and procurement under-capacity.	Ingest micro-irrigation percentage and seed replacement rate at sub-district / taluk resolution.
#3 DEOGARH Odisha (Winter 2014)	Act: 3.48 Pred: 1.34	2.14 t/ha	Hypothesis: Localized pest infestation / blast disease outbreak leading to premature lodging in lowland rice belt. Consequence: Overestimation of harvest yields results in misallocated procurement logistics and unhedged farmer credit defaults.	Integrate pest-surveillance early warning alerts and crop health telemetry into pre-harvest adjustments.
#4 SHEIKHPURA Bihar (Winter 2014)	Act: 3.81 Pred: 1.71	2.10 t/ha	Hypothesis: Extreme flood inundation causing partial harvest abandonment in riverine delta district. Consequence: Overestimation of harvest causes failure to trigger timely crop insurance payouts (PMFBY).	Fuse synthetic aperture radar (SAR) flood extent inundation masks to discount harvestable acreage.
#5 LAKHISARAI Bihar (Winter 2014)	Act: 3.69 Pred: 1.69	1.99 t/ha	Hypothesis: High agro-ecological variance across tribal/hilly tracts within the district boundary masking localized yield disparities. Consequence: Sub-optimal localized fertilizer distribution quotas based on aggregated district projections.	Refine spatial resolution to block/mandal granularity and incorporate agro-ecological zone (AEZ) stratification.

7. Operational Limitations, Inference Guardrails & Ethical Oversight

Operational Boundaries:

1. *Spatial & Temporal Support*: Predictions are valid strictly within evaluated administrative boundaries and supported temporal windows (1997–2015). Inference for unobserved districts or extrapolation beyond +4 years is rejected by ValidatedPredictor.
2. *Omitted Variable Bias*: Static regional identifiers cannot account for acute sub-seasonal meteorological shocks (cyclones, unseasonal dry spells) or sudden pest outbreaks. Feature importances reflect empirical associations, not agronomic causality.
3. *Ethical Planning Use*: Model outputs must serve as advisory decision-support for buffer stocking and regional planning. They must never be used to penalize smallholder credit access or deny crop insurance claims without physical field loss verification.
4. *Model Deserialization Security*: Only trusted bundles serialized by this verified pipeline are loaded (models/selected_bundle.joblib); deserialization of external joblib files is strictly prohibited.